Gabriel Ferreira da Costa

M.Sc. Student in Computer Science
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Research Interests

Graph Neural Networks; Spatio-Temporal Graph Learning; Time-Series Forecasting; Network Science and Complex Networks; Graph Spectral Analysis; Scientific Machine Learning; Computational Physics. ↗

The blue ↗ icon opens an official record or supporting document for the corresponding item.

Education

M.Sc. in Computer Science — Federal University of Ouro Preto (UFOP) 2025–present

Ouro Preto, Minas Gerais, Brazil

- Research title: *Benchmarking Spatio-Temporal Graph Neural Networks for Time Series Forecasting Across Heterogeneous Graph Domains.*
- Advisor: Prof. Vander Luis de Souza Freitas. Research areas: graph neural networks, complex networks, and time-series forecasting.
- Selected coursework: Algorithms; Data Mining; Neural Networks and Deep Learning; Complex Networks; Digital Image Processing.
- CAPES master's scholarship; full-time, exclusive-dedication appointment (40 hours/week). Researcher at the Intelligent Systems Computing Laboratory (CSILab/UFOP). [Transcript] [Enrollment] [Scholarship]

B.Sc. in Physics — Federal University of Ouro Preto (UFOP) 2019–2025

Ouro Preto, Minas Gerais, Brazil

- Undergraduate thesis: *Simulation of Epidemiological Models in Temporal Mobility Networks* (original title: *Simulação de Modelos Epidemiológicos em Redes de Mobilidade Temporais*).
- Advisor: Prof. Vander Luis de Souza Freitas. CNPq undergraduate research scholarship during the degree. [Degree] [Transcript]

Secondary Education — Colégio Batista de Guarulhos (CBG) [Transcript] 2016–2018

Guarulhos, São Paulo, Brazil; completed in 2018

Research Experience

Graduate Researcher and CAPES Fellow — CSILab/UFOP [Scholarship] 2025–present

Spatio-temporal graph neural networks for graph-based time-series forecasting

- Benchmarks deep-learning architectures across heterogeneous graph domains, with emphasis on spatio-temporal dependencies, graph spectral analysis, and reproducible experimental protocols.
- Advisor: Prof. Vander Luis de Souza Freitas. Related output: BRACIS 2026 full paper published in the proceedings listed below.

Undergraduate Research Fellow — CNPq / UFOP [Proof] 2024–2025

River-level forecasting with the TimesFM time-series foundation model; 20 hours/week

- Evaluated TimesFM for forecasting river levels in a basin in the Rio Negro region, with comparisons to LSTM and Prophet baselines and attention to flood- and drought-related forecasting.

Undergraduate Research Fellow — CNPq / UFOP [Proof]

2022–2024

Epidemiological processes in temporal mobility networks; 20 hours/week

- Modeled disease propagation in metapopulation networks with an Eulerian SIR formulation and temporal population inflow/outflow tensors.
- Studied synthetic mobility flows for the Belo Horizonte Metropolitan Region and 2022 IBGE census data, as well as Baidu mobility flows among 303 Chinese cities from January–February 2020; examined unrestricted and topology-constrained network scenarios.
- Related outputs: the GEOINFO 2025 full paper and the CNMAC 2024 extended abstract listed below.

Publications

Full Conference Papers

1. **Gabriel F. Costa**, Eduardo J. S. Luz, and Vander L. S. Freitas. "Benchmarking Spatio-Temporal Graph Neural Networks for Time Series Forecasting Across Heterogeneous Graph Domains." In: *Intelligent Systems*, 36th Brazilian Conference on Intelligent Systems (BRACIS 2026), *Lecture Notes in Computer Science*, vol. 17106, pp. 19–33. Springer, Cham, 2027. First published online October 2, 2026. BRACIS conference page. [Proof] [Paper]

2. **Gabriel F. Costa**, Leonardo B. L. Santos, and Vander L. S. Freitas. "Eulerian SIR Model for Asymmetric Temporal Commuting Networks: Case Study of Chinese Baidu Mobility Data." In: *Proceedings of the XXV Brazilian Symposium on Geoinformatics (GEOINFO 2025)*, vol. 25. São José dos Campos: National Institute for Space Research (INPE), 2025. Official paper. [Proof]

Extended Abstracts / Proceedings Publications

3. **Gabriel F. Costa**, Vander L. S. Freitas, and Leonardo B. L. Santos. "The SIR Metapopulation Model in a Temporal Commuting Network." *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*, 11(1), pp. 1–2, published January 20, 2025; presented at CNMAC 2024. Official publication page. [Proof]

Teaching Experience

Teaching Internship — Federal University of Ouro Preto (UFOP) [Proof]

2025–2026

BCC701 — Computer Programming I; 4 hours/week

Academic Service

Technical Program Committee Member and Reviewer [Participation] [Reviews]

2026

36th Brazilian Conference on Intelligent Systems (BRACIS 2026), Brazilian Computer Society

Technical Program Committee Member and Reviewer [Participation] [Reviews]

2026

National Meeting on Artificial and Computational Intelligence (ENIAC 2026), Brazilian Computer Society

Undergraduate Thesis Examination Committee Member [Declaration]

2026

B.Sc. in Computer Science, Federal University of Ouro Preto

Committee for Jéssica Giovana Zacarias, *Analysis of the Curriculum Structure of the Federal University of Ouro Preto through Academic Prerequisite Networks* (original title: *Análise da Estrutura Curricular da Universidade Federal de Ouro Preto por Meio de Redes de Pré-Requisitos Acadêmicos*).

Academic Profiles

ORCID	0009-0001-2714-3365 ↗
Lattes CV	lattes.cnpq.br/6533107458311261 ↗
GitHub	github.com/gabrielxcosta ↗
LinkedIn	linkedin.com/in/gabxcostaxf ↗
Google Scholar	Google Scholar profile ↗
Bibliographic name	COSTA, G. F. ↗
Evidence repository	Google Drive folder ↗

Languages

Portuguese Native ↗
English EF SET: 76/100, C2 Proficient (Reading: 80; Listening: 71), awarded February 17, 2025. [Certificate]

Conference Presentations and Academic Events

2026 — ENREDANDO ↗

Poster and lightning talk, "Benchmarking Spatio-Temporal Graph Neural Networks for Time Series Forecasting Across Heterogeneous Graph Domains," Ibero-American School on Networks and Complex Systems, Viçosa, Brazil, July 26–31, 2026. [Presentation] [Certificate]

2026 — BRACIS 2026

Oral presentation of "Benchmarking Spatio-Temporal Graph Neural Networks for Time Series Forecasting Across Heterogeneous Graph Domains," 36th Brazilian Conference on Intelligent Systems, Cuiabá, Mato Grosso, Brazil, October 19–22, 2026. Conference participation: 40 hours. [Presentation] [Certificate]

2025 — GEOINFO [Certificate]

Symposium participation associated with "Eulerian SIR Model for Asymmetric Temporal Commuting Networks: Case Study of Chinese Baidu Mobility Data," XXV Brazilian Symposium on Geoinformatics, São José dos Campos, Brazil.

2024 — CNMAC [Certificate]

Poster, "The SIR Metapopulation Model in a Temporal Commuting Network," National Congress of Applied and Computational Mathematics, Ipojuca, Brazil.

2024 — Encontro de Saberes UFOP [Authorship] [Certificate]

Poster, "Simulation of Epidemiological Models in Temporal Mobility Networks."

2023 — Brazilian Physics Society Autumn Meeting [Presentation] [Certificate]

Presentation, "The SIR metapopulation model in a temporal commuting network," session *Complex Networks, Econophysics and Sociophysics II*, Ouro Preto, Minas Gerais, Brazil, May 21–25, 2023. Authors: **Gabriel F. Costa**, Matheus M. G. Correia, Leonardo B. L. Santos, and Vander L. S. Freitas.

2023 — Encontro de Saberes UFOP [Authorship] [Certificate]

Poster, "Simulation of Epidemiological Models in Temporal Mobility Networks."

Additional Academic Training

II School on Modelling Infectious Disease Dynamics [Certificate] 2025

Institute for Theoretical Physics Foundation (IFT), São Paulo, Brazil — short course, 40 hours

GeoPandas: Working with Geospatial Data [Certificate] 2023

Alura — online course; original title: Geopandas: trabalhando com dados geoespaciais*; completed October 19, 2023; 6 hours*

Introduction to Machine Learning Applied to Physics [Certificate] 2023

Brazilian Physics Society Autumn Meeting, Ouro Preto, Minas Gerais, Brazil — tutorial taught by Henrique Ferreira, 3 hours

Python for Data Science — Training Program [Certificate] 2022

Alura — online professional training; official title: Formação Python para Data Science*; completed December 27, 2022; estimated workload: 86 hours*

Web Scraping with Python: Collecting Data from the Web [Certificate] 2022

Alura — online course; original title: Scraping com Python: coleta de dados na web*; completed December 27, 2022; 10 hours*

Statistics with Python — Training Program [Certificate] 2022

Alura — online professional training; official title: Formação Estatística com Python*; completed November 29, 2022; estimated workload: 71 hours*

Technical Skills

Programming Languages and Workflows	Python; Snakemake; C; C++; Bash ↗
Machine Learning and Graph Learning	PyTorch; PyTorch Geometric; PyTorch Geometric Temporal; Scikit-learn; Graph Neural Networks; Spatio-Temporal Graph Neural Networks; graph spectral analysis ↗
Scientific Computing and Data Analysis	NumPy; SciPy; Pandas; Matplotlib; Jupyter; Python multiprocessing; statistical data analysis; web scraping ↗
Geospatial and Network Analysis	GeoPandas; Shapely; OSMnx ↗
Research and Development Tools	Git; Linux; Docker; Slurm; LaTeX; Markdown; VS Code ↗

Selected Public Repositories

Spatiotemporal GNN Forecasting ↗
Research implementation and experiments investigating spatial and temporal modules in graph neural networks for time-series prediction on static and dynamic graphs.

Epidemiological Models in Temporal Mobility Networks ↗
Research code and materials for comparing epidemiological processes on temporal mobility networks and aggregated static representations.

Complex Networks ↗
Implementations and study materials covering assortativity, community detection, scale-free and random networks, graph robustness, cascading failures, and graph theory.

Leadership and Institutional Activities

Communication Board Advisor — Association of Federal Student Houses of Ouro Preto (REFOP) [Certificate]
2019–2020; 2021–2022

Federal University of Ouro Preto (UFOP), Ouro Preto, Minas Gerais, Brazil

Served as an advisor to the Communication Board across two administrative terms: August 1, 2019–August 1, 2020 (20 hours), and August 1, 2021–August 1, 2022 (100 hours); 120 hours total.

Additional Professional Experience

Freelance Waiter
Ouro Preto, Minas Gerais, Brazil

Seu José; Tenente Pimenta; Recanto Serra do Trovão.